

DP Computer Science: A 2.1.2 Modern Digital Infrastructures

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the **purpose** and **benefits** of modern digital infrastructures
 - Identify key components like the **internet**, **cloud computing**, and **distributed systems**
 - Analyse the impact of digital infrastructures on **society** and the **environment**
 - Think critically about the **ethical implications** of technology
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Key Vocabulary

Term	Definition
Digital infrastructure	The underlying systems that enable digital communication, data storage, and services
Internet	The global network connecting billions of devices worldwide
Cloud computing	Delivering computing services (servers, storage, software) over the internet
Distributed system	A system where components are spread across multiple locations and communicate over a network
Edge computing	Processing data closer to where it is generated, reducing latency
Mobile networks	Wireless networks that connect mobile devices (3G, 4G, 5G)

Term	Definition
WWW (World Wide Web)	A system of interlinked documents and resources accessed via the internet
Blockchain	A distributed ledger technology that records transactions securely and transparently
Digital divide	The gap between those who have access to digital technologies and those who do not

IB ATL Skills

Skill	Description
Communication	Communicating information and ideas effectively
Collaboration	Working collaboratively with peers
Thinking	Applying thinking skills critically and creatively
Social	Demonstrating empathy and respect for different perspectives

2. Purpose of Modern Digital Infrastructures

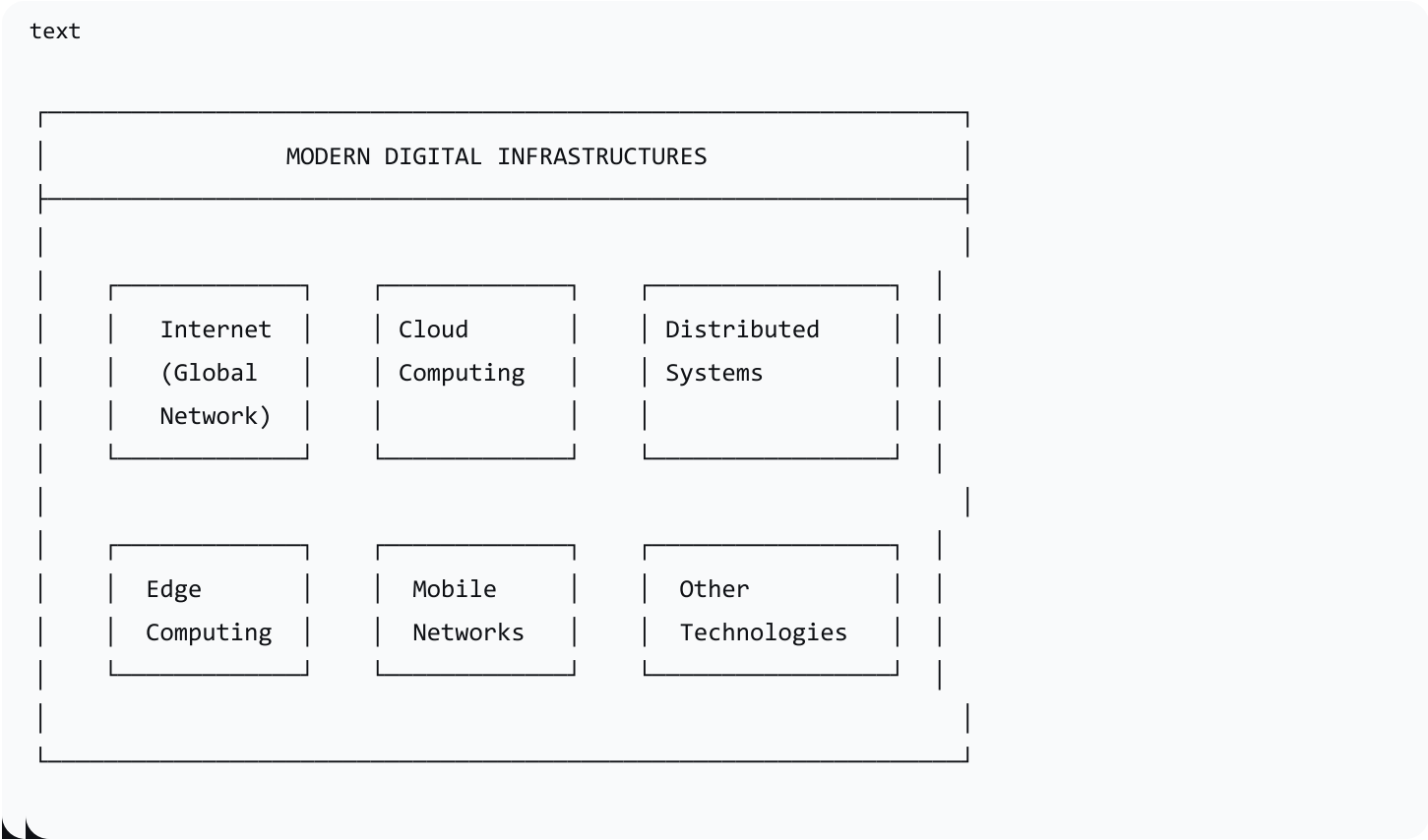
Core Purposes

Modern digital infrastructures serve four main purposes:

Purpose	Description
Connectivity	Connecting people, devices, and systems across the globe
Data Handling	Storing, processing, and transmitting vast amounts of data

Purpose	Description
Service Delivery	Delivering digital services (e.g., streaming, banking, e-commerce)
Innovation	Enabling new technologies and business models

Visual Overview



3. Key Components of Modern Digital Infrastructures

Component 1: The Internet

Aspect	Description
Definition	The global network connecting billions of devices worldwide

Aspect	Description
Role	The foundation of most digital services; enables communication and data exchange
Key Features	Global reach, decentralised, based on TCP/IP protocols
Applications	Web browsing, email, streaming, social media, IoT

Component 2: Cloud Computing

Aspect	Description
Definition	Delivering computing services (servers, storage, software) over the internet
Role	Provides scalable, on-demand computing resources
Benefits	Cost-effective, scalable, accessible from anywhere
Limitations	Relies on internet connectivity, data security concerns, latency
Examples	AWS, Google Cloud, Microsoft Azure, Dropbox, Google Drive

Cloud Service Models:

Model	Description	Example
SaaS	Software delivered over the internet	Gmail, Google Docs
PaaS	Platform for developing applications	Google App Engine
IaaS	Fundamental computing resources (servers, storage)	AWS EC2

Component 3: Distributed Systems

Aspect	Description
Definition	A system where components are spread across multiple locations and communicate over a network
Role	Distributes processing and storage across multiple machines
Benefits	Fault-tolerant, scalable, can handle massive workloads
Limitations	Complexity, coordination overhead, network dependence
Examples	Blockchain, content delivery networks (CDNs), distributed databases

Component 4: Edge Computing

Aspect	Description
Definition	Processing data closer to where it is generated, rather than in a centralised cloud
Role	Reduces latency, decreases bandwidth usage, improves responsiveness
Benefits	Faster response times, lower data transmission costs
Limitations	Limited processing power at the edge, security concerns
Examples	Smart cameras, autonomous vehicles, IoT devices

Cloud vs. Edge Computing:

Feature	Cloud Computing	Edge Computing
Processing Location	Centralised data centres	Near the data source
Latency	Higher	Lower

Feature	Cloud Computing	Edge Computing
Bandwidth	Requires more bandwidth	Reduces bandwidth usage
Scalability	Highly scalable	Limited by edge devices
Example	Large-scale data processing	Real-time analytics, autonomous vehicles

Component 5: Mobile Networks

Aspect	Description
Definition	Wireless networks that connect mobile devices
Role	Provides connectivity for smartphones, tablets, and IoT devices
Generations	3G, 4G, 5G (increasing speed and capability)
Benefits	Mobility, widespread coverage
Limitations	Coverage gaps, data caps, infrastructure cost
5G Features	High speed, low latency, massive device connectivity

4. Applications of Digital Infrastructures

Application	Description	Infrastructure Components
World Wide Web (WWW)	System of interlinked documents and resources accessed via the internet	Internet, cloud computing
Blockchain	Distributed ledger technology for secure, transparent record-	Distributed systems, internet

Application	Description	Infrastructure Components
	keeping	
Smart Traffic Systems	Adaptive traffic lights using real-time data to reduce congestion	Edge computing, IoT, mobile networks
School Networks	Connecting students, teachers, and resources within an educational institution	LAN, internet, cloud computing

5. Benefits of Digital Infrastructures

Benefit	Description
Efficiency	Automates processes, reduces manual work, speeds up operations
Access to Information	Global access to knowledge and data
Economic Growth	Enables new business models, creates jobs, drives innovation
Connectivity	Connects people across the globe, enabling communication and collaboration
Scalability	Systems can grow with demand
Cost-Effectiveness	Reduces costs through automation and resource sharing

6. Limitations and Challenges

Limitation	Description
Cost	Expensive to build, maintain, and upgrade infrastructure
Security Risks	Cyberattacks, data breaches, privacy concerns
Digital Divide	Gap between those with access to digital technologies and those without
Reliance on Connectivity	Services require stable internet access
Energy Consumption	Data centres consume significant electricity
E-Waste	Disposal of obsolete hardware

The Digital Divide

Aspect	Description
Definition	The gap between those who have access to digital technologies and those who do not
Causes	Income inequality, geographic location, lack of digital literacy
Impacts	Unequal access to education, employment, healthcare, and services
Solutions	Investment in infrastructure, digital literacy programs, affordable devices

7. Societal and Environmental Impacts

Societal Impacts

Positive Impacts	Negative Impacts
Improved access to education and information	Disinformation and misinformation
Enhanced communication and collaboration	Social isolation
Economic opportunities and innovation	Job displacement through automation
Access to healthcare services (telemedicine)	Inequality (digital divide)
Democratic participation and engagement	Privacy concerns and surveillance

Environmental Impacts

Impact	Description
Energy Consumption	Data centres and networks consume large amounts of electricity
E-Waste	Discarded electronics contribute to pollution and resource depletion
Potential Solutions	Using renewable energy for data centres, more efficient hardware, circular economy approaches
Sustainable Development	Digital technologies can help monitor and address environmental challenges

8. Ethical Considerations

Ethical Issue	Description
Data Privacy	Who owns data? How is it collected, stored, and used?
AI Bias	Artificial intelligence systems may reproduce or amplify existing biases

Ethical Issue	Description
Responsible Tech Use	Ensuring technology is developed and used for the benefit of all
Surveillance	Balancing security needs with individual privacy rights
Autonomy	Human control over automated systems and decision-making

Key Questions for Discussion

Question	Purpose
"How can we minimise the negative impacts of digital infrastructures?"	Encourages problem-solving and critical thinking
"What are the ethical dilemmas surrounding data privacy and AI?"	Develops awareness of ethical issues
"What are potential solutions to the digital divide?"	Encourages thinking about social responsibility

9. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	5 min	"Tech in Your Life" brainstorm—how technology impacts daily life
Explore	15 min	Purpose and benefits of digital infrastructures; Think-Pair-Share on benefits and drawbacks

Phase	Time	Activity
Explain	20 min	Key components (internet, cloud, distributed systems, edge computing, mobile networks); Interactive quiz
Elaborate	15 min	Impact analysis: groups assigned to societal, environmental, or ethical aspects
Wrap-up	5 min	Key takeaways; exit ticket

Impact Analysis Activity

Task: In groups, analyse one of the following aspects:

Group	Aspect	Guiding Questions
Group 1	Societal Impacts	How do digital infrastructures affect communities? What are the positive and negative effects?
Group 2	Environmental Impacts	What is the environmental cost of digital infrastructures? How can they contribute to sustainability?
Group 3	Ethical Considerations	What are the key ethical dilemmas? How can we ensure responsible tech use?

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions during discussions

Assessment Type	Description
Interactive Quiz	Use online tools (Quizlet) to assess understanding of key components
Impact Analysis Activity	Assess ability to analyse impacts and consider ethical implications
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Worksheet Questions:

1. What are the purposes of modern digital infrastructures?
2. Briefly describe the role of:
 - Cloud computing
 - Edge computing
3. What is one potential societal and environmental impact of digital infrastructures?
4. Give an example of an ethical dilemma related to technology.

11. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to delve deeper into specific technologies or ethical issues

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.1.2 with quiz)
- **Worksheet:** Purpose, components, and impact questions
- **Quizlet** (A2.1.2 vocabulary flashcards)

13. Summary: Key Takeaways

Concept	Key Point
Purpose	Connectivity, data handling, service delivery, innovation
Internet	Global foundation for digital services
Cloud Computing	On-demand computing resources over the internet
Edge Computing	Processing data closer to the source (low latency)
Distributed Systems	Components spread across multiple locations (fault-tolerant)
Digital Divide	Gap between those with and without access to technology
Ethical Issues	Data privacy, AI bias, surveillance, responsible technology use
Environmental Impact	Energy consumption, e-waste, potential for sustainable solutions

"Digital infrastructures are the backbone of modern society—they enable communication, innovation, and economic growth, but they also bring challenges that require careful, ethical, and sustainable development."